

Using Normal-Model z-Scores and Percentiles

Standardizing, unstandardizing, the empirical rule, and comparing across distributions

Directions:

- Standardize with $z = \frac{x - \mu}{\sigma}$; go back with $x = \mu + z\sigma$.
- Every distribution on this sheet is assumed to be approximately normal.
- Percentages come from the empirical rule: about 68% of values lie within 1 standard deviation of the mean, about 95% within 2, and about 99.7% within 3.
- Give z -scores to two decimal places where they are not exact.

1. An exam has mean $\mu = 500$ and standard deviation $\sigma = 100$. Find the z -score of a score of 620.

Answer: _____

2. On the same exam, find the z -score of a score of 455.

Answer: _____

3. On the same exam, what percentage of scores fall below 600? (Use the empirical rule; give a whole percentage.)

Answer: _____

4. On the same exam, a student's z -score is 1.5. Find their exam score.

Answer: _____

5. A student scores 88 on a history test with $\mu = 80$ and $\sigma = 5$, and 76 on a biology test with $\mu = 70$ and $\sigma = 4$.

- (a) Find the z -score on the history test. _____
- (b) Find the z -score on the biology test. _____
- (c) Explain which performance was stronger relative to its class, and why the raw scores alone cannot settle it.

6. Adult heights in a population are approximately normal with $\mu = 170$ cm and $\sigma = 8$ cm. About 95% of heights lie between two values.

- (a) Find the lower of those two heights.

Answer: _____ cm

- (b) Find the upper of those two heights.

Answer: _____ cm

7. For the same heights ($\mu = 170$ cm, $\sigma = 8$ cm), find the z -score of a height of 182 cm.

Answer: _____

8. For the same heights, what percentage of the population is shorter than 162 cm? (Use the empirical rule; give a whole percentage.)

Answer: _____

9. A battery's life is approximately normal with $\mu = 40$ hours and $\sigma = 3$ hours. Find the 97.5th percentile of battery life — the value with 97.5% of batteries below it.

Answer: _____ hours

10. For the same batteries, one is chosen at random. Using the empirical rule, 25 batteries in every 1000 last longer than the 97.5th percentile. Give that probability as a fraction in lowest terms.

Answer: _____

11. On a test with $\mu = 72$ and $\sigma = 6$, a student's z -score is -1.5 .

- (a) Find the student's raw score. _____
- (b) Explain what the negative sign tells you, and estimate the student's percentile from the empirical rule.

12. Synthesis. Line 1 produces batteries with $\mu = 40$ hours and $\sigma = 3$ hours; Line 2 produces them with $\mu = 45$ hours and $\sigma = 5$ hours. One battery from Line 1 lasted 46 hours; one from Line 2 lasted 52 hours.

- (a) Find the z -score of the Line 1 battery. _____
- (b) Find the z -score of the Line 2 battery. _____
- (c) Explain which result is more unusual for its own production line, even though one battery lasted longer than the other.